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Co-Producing Coastal Bird Research on the Gulf Coast through Structured Decision Making (SDM) and Constructed Value of Information (CVoI)

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Abstract
Conservation practitioners face the complex challenge of addressing global biodiversity threats within social-ecological systems with multiple jurisdictions, numerous interest groups, and changing environmental conditions. Bridging the research-to-implementation gap between scientists and managers, which integrates diverse perspectives that represent the variety of considerations impacting management outcomes can help to inform efficient and effective conservation. Avian conservation programs along the U.S. Gulf Coast exemplify these challenges, as management decisions must address diverse threats to declining coastal bird populations across large breeding, wintering, and migratory ranges while facing uncertainty surrounding the efficacy of large-scale strategies. This provides opportunities for improving management through co-produced research with structured decision making (SDM). We present a case study that demonstrates the application of co-production principles and SDM to collaboratively develop high-priority research questions informing effective avian conservation and management. Additionally, we applied a constructed value of information (CVoI) framework to identify and prioritize which uncertainties to reduce and better understand how research questions impact management actions. The integration of SDM's five-step decision making framework ensured key principles of co-production were met, while co-production enhanced SDM outcomes by incorporating diverse perspectives and knowledge bases. CVoI further allowed participants to collectively determine final high-priority research questions in the face of complexity and uncertainty. Together, these approaches supported co-produced knowledge-generation, strategic hypothesis prioritization, and identification of research questions most likely to improve conservation decisions. This case study from the U.S. Gulf Coast demonstrates how co-produced research through SDM offers a broadly applicable approach for navigating ecological complexity and bridging the research-to-implementation gap in avian conservation.

Keywords Co-Production · Decision Making · Research-to-Implementation Gap · Actionable Science · Conservation · U.S. Gulf Coast · Social Science · Management

Introduction

Across the globe, researchers, land managers, and conservation practitioners are confronted with the challenge of strategically allocating limited resources to address large-scale, complex problems (Mace et al., 2000; Margules & Pressey, 2000; Hostetler et al., 2015; Fournier et al., 2023). The design and implementation of effective conservation strategies are rarely straightforward due to the multifaceted and interconnected nature of threats, the involvement of diverse groups with differing goals, persistent uncertainties about ecological responses, and the inherent unpredictability of social-ecological systems (Beier et al., 2017; Norström et al., 2020; Stantial et al., 2023). Because these challenges

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Extended author information available on the last page of the article cross both geographic and political boundaries and demand diverse forms of expertise (e.g., academic, managerial), co-producing knowledge among science producers and users can help achieve tangible conservation outcomes and overcome what is often called the research-to-implementation gap (Armitage et al., 2011; Beier et al., 2017; Byrd et al., 2023; Grimm et al., 2025). Here, we define *science producers* as ecological researchers and participatory scientists (i.e., citizen scientists), and *science users* as decision makers, resource managers, and policy makers (*sensu* Beier et al., 2017; Saunders et al., 2021).

Bridging the research-to-implementation gap involves the transfer of actionable science between science producers and users, where actionable science is defined as scientific findings (e.g., tools, data, analyses) that inform management decisions (sensu Beier et al., 2017). Traditionally, however, knowledge transfer has often relied on a linear transfer of knowledge between research findings and implementation, in which science producers are removed from decision-making under a "pure scientist" or "science arbiter" role (Pielke, 2007). In this approach, science producers may expect users to independently pick up and implement their research findings from a "loading dock" (e.g., in a journal article, report) (Cash et al., 2006; Beier et al., 2017). However, this model often fails when faced with uncertainty and conflicting values and/or objectives, resulting in a "research-to-implementation gap" where scientific findings are generated in isolation from specific decision-making contexts, resulting in knowledge that is underutilized or difficult to apply (Saunders et al., 2021; Pérez-Granados & López-Iborra, 2022; Fournier et al., 2023).

Alternatively, science producers can adopt a more involved role such as the "honest broker of policy alternatives," aligning with the "stakeholder model" view in which science users are involved in science production and the role of science in decision making is considered (Pielke, 2007). This is most effective when a wide range of perspectives and knowledge are supported and integrated by following the principles and guidance of well-established collaborative practices (e.g., co-production, two-way knowledge exchange, social learning, contractual research, partnering with boundary organizations; Cash et al., 2003; Cash et al., 2006; Cook et al., 2013; Kirchhoff et al., 2013; Meadow et al., 2015). To close the research-to-implementation gap, these engagement practices are best paired with or already embedded within decision analyses that formally allow diverse participants to collaboratively and transparently frame objectives and evaluate alternatives. A wide variety of these frameworks have been developed and utilized by U.S. public agencies, such as systematic conservation planning (Margules & Pressey, 2000), evidence-based practice (Sutherland et al., 2004), adaptive management (Walters, 1986), and among the most commonly used terms in natural resource management, structured decision making (SDM; Gregory et al., 2012). Structured decision making includes a variety of tools for choosing among alternative actions to address a problem, and when problems include choosing which uncertainty is most informative to reduce for management decisions, constructed value of information can be useful for including end users and experts in selecting high impact uncertainties for future research and learning (CVoI; Runge et al., 2023).

Knowledge co-production through the implementation of SDM can aid avian conservation efforts, where the expansive and dynamic nature of species' ranges and habitat use results in numerous threats to address across numerous jurisdictions with varied priorities and values. Similar to conservation efforts broadly (e.g., Cook et al., 2013; Fabian et al., 2019; Ausden & Walsh, 2020), avian conservation also faces research-to-implementation gaps that can hinder program efficacy (e.g., Walsh et al., 2015; Pérez-Granados & López-Iborra, 2022; Livingston et al., 2024). The U.S. Gulf Coast, for example, is a high-priority region for implementing avian conservation (Wilson et al., 2019), yet efforts are limited by challenges such as finite resources, science products and study designs that do not align with management priorities, logistical complexities, difficulties identifying and coordinating across interest groups, and insufficient two-way communication among science producers and users (Wilson et al., 2019; Fuller et al., 2025; Fournier et al., 2023; National Academies of Sciences, Engineering, and Medicine, 2018). Effectively addressing such issues necessitates evidence-based management coordinated across broad scales and disciplines (Saunders et al., 2021). This paper provides a case study along the northern Gulf Coast of the United States (hereafter, Gulf Coast) demonstrating how co-produced research through SDM can help overcome this gap in avian conservation.

Co-Production

Co-production is a well-established process that can be used to address the research-to-implementation gap. Other terms for similar inclusive approaches to science production exist (e.g., translational ecology; Enquist et al., 2017; Saunders et al., 2021), but all emphasize promoting active participation and fostering partnerships across science producers and users to develop actionable science (Schlesinger, 2010; Brunson & Baker, 2016; Beier et al., 2017; Saunders et al., 2021). For clarity and simplicity, we will focus on the application of co-production as this is commonly used within management settings. When applied effectively, processes like co-production identify priorities and build capacity and networks for implementing conservation and natural resource management decisions (Hastings et al., 2020; Norström et al., 2020). Effective co-production relies on diverse contributions, where science producers provide expertise in research development; managers share their expertise and identify research needs and how to apply findings; and interest groups (e.g., affected communities, potential resource users) help to highlight constraints, alternatives, and management implications (Beier et al., 2017).

Multiple guiding principles and recommendations are outlined in the literature for carrying out the co-production process (Reed et al., 2014; Meadow et al., 2016; Beier et al., 2017; Djenontin & Meadow, 2018; Norström et al., 2020). For clarity, this paper focuses on how our work follows the co-production principles outlined by Norström et al. (2020): namely, that the process should be rooted in a specific problem or set of problems ("context-based"), embrace multiple ways of knowing and doing ("pluralistic"), focus on a clearly defined goal that is important to all interest groups ("goal-oriented"), and foster ongoing, active collaboration throughout the process ("interactive"). In addition to these principles, we acknowledge and have incorporated some of the approaches suggested for addressing the challenge of engaging participants in the full co-production process (e.g., Meadow et al., 2015; Oliver et al., 2019, Beier et al., 2017). For example, we included the formation of partner organizations (e.g., technical advisory groups and steering committees) to help distribute some of the roles and responsibilities (e.g., goal adjustments, reviewing key decisions, managing timelines and workflows; Beier et al., 2017).

Engaging with and incorporating contributions from diverse participants alone is insufficient without intentionally fostering trust. This can be achieved by facilitating transparent and genuine exchange of knowledge and recognizing power dynamics. Co-production aims to build trust and engagement among participants through sustained interactions, inclusion of science users throughout the process, and commitment to the process and the group (Lemos & Morehouse, 2005; Enquist et al., 2017; Djenontin & Meadow, 2018; Norström et al., 2020), which in turn is more likely to lead to successful, research-informed implementation (Lemos & Morehouse, 2005). In addition, open communication and two-way knowledge exchange, where science producers also learn from information shared by users (Cash et al., 2006), are key for bridging the gap between knowledge and practice in conservation (Byrd et al., 2023). Finally, those engaging in co-production processes should be aware of how their positions and potential power can influence how participants are selected, how knowledge is prioritized, and the ways different people can participate (Turnhout et al., 2020; Gerlak et al., 2023).

Structured Decision Making

Co-production lays the foundation for collaborative knowledge generation but does not necessarily provide a direct, structured pathway for translating knowledge into implementation. Meadow et al. (2015) notes that research on co-production has primarily focused on what the process might involve (e.g., diverse partnerships, sustained collaboration) rather than mechanisms to implement these goals. For example, prior to 2015, only 206 papers in Google Scholar (Alphabet Inc., 2025) reference "co-production" in connection with implementation mechanism terminology such as "decision analytic framework", "multi-criteria decision analysis", or "structured decision making". Although this number has grown to around 1,700 as of early 2026, it remains negligible compared to over 1,000,000 papers referencing "co-production" without these aforementioned terms. This highlights the relatively limited attention given to structured implementation mechanisms within the study of co-production.

Conversely, practitioners applying structured decision making (SDM) analytic frameworks have included guidelines for effective collaboration that, while not always explicitly referred to as "co-production", share similar principles (Nicholson et al. 2002; Starfield, 1997). Similarly, our work included the formalized principles of co-production as a complementary component of SDM, which in turn provides a mechanism to put co-produced knowledge into practice. SDM, defined here as a framework for evaluating complex decisions by breaking them down into parts (sensu Robinson & Fuller, 2017), provides a systematic approach for identifying, prioritizing, and addressing critical uncertainties related to complex ecological problems. SDM enables science producers and users to collaboratively, transparently, and thoroughly generate knowledge, prioritize objectives, and implement coordinated, well-informed decisions (Gregory et al., 2012; Fournier et al., 2019, 2020; Wright et al., 2020; Stantial et al., 2023).

SDM employs a systematic method of defining the **Problem**, **Objectives**, and **Alternatives**, then evaluating **Consequences** and **Tradeoffs** (PrOACT; Hammond et al., 1998; Gregory et al., 2012; Fournier et al., 2023). Problem framing generates a common understanding and definition of the problem the group seeks to address. After framing the problem, participants collectively determine the fundamental objective (i.e., end goal or core value) as well as the means objective that illustrates how the fundamental objective can be achieved through an objective hierarchy (Keeney, 1992; Gregory et al., 2012; Rushing et al., 2020; Fournier et al., 2023). Participants then identify management alternatives and connect them to means and fundamental objectives with relevant ecological factors and biological responses

(Gregory et al., 2012). These can then be used to assess tradeoffs among alternative actions, which can be approached in a variety of ways, including through influence diagrams, allowing contributors to visually link alternatives to objectives and outcomes, thereby generating a series of potential testable mechanisms (Howard & Matheson, 2005; Robinson & Fuller, 2017). Influence diagrams demonstrate the mechanisms by which each management alternative may impact objectives and set the stage to evaluate consequences (i.e., impacts of each alternative) and tradeoffs (i.e., comparative strengths and weaknesses; Keeney, 2004; Gregory et al., 2012).

Constructed Value of Information

Within the SDM framework, participants evaluate how uncertainty influences decisions. Uncertainty often makes it difficult to effectively identify the optimal decision alternative based on the values and needs of those involved. Therefore, it is important to understand which sources of uncertainty, if reduced, would influence the selection of an alternative that most improves the desired outcome. In adaptive management, decision-relevant uncertainty can often be reduced through targeted research and, in turn, affect which management actions are implemented (Runge et al., 2011).

Value of Information (VoI) is an additional decision analytic framework, which provides a formal method to assess how much reducing uncertainty or gaining knowledge, will improve decision making (Bolam et al., 2019; Runge et al., 2020, 2023; Lawson et al., 2022). When quantitative data required for VoI is not readily available and/or when it is early in the decision-making process, Constructed Value of Information (CVoI; formerly Qualitative Value of Information) can provide a solution (Runge et al., 2023). CVoI involves experts (including both science producers and users) scoring alternatives according to three criteria: (1) *Magnitude*, or impact of uncertainty on the objective outcome, (2) *Relevance*, or how pertinent the uncertainty is to decision making, and (3) *Reducibility*, or how much the uncertainty can be decreased. CVoI, the product of Magnitude and Relevance, is then compared to Reducibility to identify options that are Pareto optimal (Williams & Kendall, 2017; Runge et al., 2020; Rushing et al., 2020; Lawson et al., 2022). Pareto optimal alternatives aim to maximize CVoI and Reducibility, where choosing another alternative would reduce both attributes. While this ensures efficiency, it does not guarantee equitable outcomes as the decision maker may be required to compromise on either CVoI or Reducibility (Williams & Kendall, 2017; Stantial et al., 2023).

Although VoI approaches and metrics (e.g., expected value of perfect information (EVPI), CVoI) are increasingly being applied within the SDM framework, including several ecological applications (Johnson et al., 2017; Lawson et al., 2022; Sipe, 2023; Stantial et al., 2023; Gerber et al., 2024; Rushing et al., 2020; Runge et al., 2023; Martin & Johnson, 2019; Martin et al., 2025) few if any include social science alternatives. Despite this, there is a growing recognition of the importance of integrating the social sciences to better inform management and more effectively address conservation issues, highlighting the importance of considering social, economic, cultural, and jurisdictional contexts during planning and improving the equitability and inclusivity of conservation processes (Bennett et al., 2017; Dayer et al., 2020; Niemiec et al., 2021; Robinson et al., 2019). Furthermore, social science-based management efforts have proven beneficial at large, coordinated scales (Reid et al., 2016; Dayer et al., 2020; Manfredo et al., 2021; Pitman et al., 2021). However, a major barrier to integrating social science approaches is the failure to include social science throughout the decision-making process (Niemiec et al., 2021; Robinson et al., 2019). SDM protocols emphasize that all areas of expertise involved in a decision are needed in the process, and our study applies multiple recommendations from Robinson et al. (2019) to advance social science integration by incorporating social science-based alternatives throughout the process, applying CVoI to evaluate tradeoffs and objectives through structured surveys, and presenting a reference case study of how this process can be accomplished in a management setting.

Here, we present a case study highlighting the use of SDM techniques in an inclusive co-production process to collaboratively design, carry out, and implement a study evaluating human disturbance impacts on coastal breeding birds of the Gulf Coast. Through the presentation and analysis of this case study, we seek to meet two objectives: evaluating the ability of co-production and SDM techniques to address the research-to-implementation gap in large-scale avian conservation and evaluating the effectiveness of tools such as CVoI in facilitating conservation decision making.

Methods

Study Context: Gulf Coast and the Gulf Avian Monitoring Network

Coordination across geographies and jurisdictions is especially critical in biodiversity-rich regions like the Gulf Coast, where conservation is challenged by numerous concurrent threats, uncertainty regarding threat prioritization and effective mitigation strategies, and diversity of priorities among interest groups (Wilson et al., 2019; Fuller et al., 2025). The region supports at least 395 bird species during their annual life cycle and provides critical breeding and non-breeding habitat of continental and hemispheric importance for many coastal birds (e.g., shorebirds, seabirds, and wading birds; Withers, 2002; Gallardo et al., 2009; Remsen et al., 2019; Wilson et al., 2019). Many of these coastal bird species have experienced sharp population declines both across North America and globally, necessitating urgency in maintaining suitable habitat (Rosenberg et al., 2019; Wilson et al., 2019; Smith et al., 2023). Gulf Coast birds face a variety of threats, including human disturbance, habitat loss and/or degradation, unbalanced predator populations, extreme weather events, and exposure to contaminants including microplastics (Wilson et al., 2019; Johnson, 2021; Grace et al., 2022, 2024; Fuller et al., 2025; Zenzal et al. 2025). Human disturbance in the region is amplified by heavy recreational shoreline use (Brush et al., 2019; Jodice et al., 2019; Fuller et al., 2025), while habitat loss and/or degradation result from both climatic and geologic events (e.g., erosion, severe weather, sediment loss), and human-driven causes (e.g., urban development, anthropogenic climate change, chemical spills and runoff; Morton, 2003; Mendelssohn et al., 2017; Zenzal et al. 2025).

In addition to substantial uncertainty surrounding effective strategies to address and prioritize well-documented threats in the Gulf Coast (but see Mengak et al., 2019), obstacles to conservation actions include resource constraints and lack of large-scale coordination (Wilson et al., 2019; Fuller et al., 2025; National Academies of Sciences, Engineering, and Medicine, 2018). Fragmented research in the region has resulted in inconsistent study designs, objectives, hypotheses, monitoring protocols, and spatial and temporal scales (Fournier et al., 2019, 2020, 2023). These challenges can in part be attributed to the difficulty of coordination among diverse partner agencies and organizations with varying priorities, limited funding to support large scale projects, as well as pressure from competing human interests (e.g., recreation, development) and wildlife needs on limited shoreline habitat.

For the reasons stated above, the northern Gulf Coast is well-suited for: co-production to coordinate collaboration across a wide variety of partners and perspectives, utilizing SDM frameworks to provide a structured method for translating knowledge into actionable science, and tools such as CVoI to identify research questions most critical for reducing uncertainty in how management actions will address conservation objectives. The first step to collaboratively address these concerns among science producers and users was the formation of the Gulf Avian Monitoring Network after the Deepwater Horizon oil spill in 2010 (DHNRDAT, 2016; Wilson et al., 2019). The Gulf Avian Monitoring Network, a co-produced, SDM-based effort involving science producers and users from all U.S. Gulf Coast states (Texas, Louisiana, Mississippi, Alabama, and Florida), identified primary threats, management recommendations, and future monitoring priorities of birds in the region (Wilson et al., 2019; Fournier et al., 2023). As a continuation of the Gulf Avian Monitoring Network's work and guided by its research framework (Wilson et al., 2019), our team identified uncertainties and designed a targeted research study to reduce priority uncertainties and improve decision making for coastal bird stewardship and management.

Co-Production Team Development and Literature Review

In response to a funding opportunity from the National Oceanic and Atmospheric Administration's (NOAA) RESTORE Science Program in 2020, we formed a co-production team of 12 Co-Principal Investigators (Co-PIs; including eight science producers and four science users) and nine Technical Advisory Group (TAG) members (including one science producer and eight science users; Table 1). This team drew upon their diverse conservation and management expertise and the Gulf Avian Monitoring Network's evaluation of

Table 1 Workshop participant summary ($n=33$), including the participants' associated organization, science role (i.e., science producer or user), number of participants (i.e., individuals from that organization category and science role who attended one or more co-production meetings or workshops), number of co-PIs (on the NOAA RESTORE co-production award), and number of TAG members/collaborators

Organization Category	Science Producer/User	Number of Participants	Number of Co-PIs	Number of TAG members/collaborators
Academic	Producer	5	4	0
Federal Agency	Producer	2	1	1
Federal Agency	User	3	1	1
Independent Consultant	Producer	1	0	1
Large-scale NGO	Producer	6	3	0
Large-scale NGO	User	3	0	3
Local Agency	User	1	0	0
Private Organization	User	2	1	0
Regional NGO	User	6	2	3
Total		33	12	10

the threats and uncertainties affecting Gulf Coast bird and habitat management to prepare a proposal - later awarded - focused on coastal bird stewardship and management, as these had been identified as two of the greatest threats to Gulf Coast birds (Wilson et al., 2019).

The co-production team met regularly via monthly calls and workshops. Co-PIs committed to full participation throughout the process, while TAG members and additional collaborators participated as available. Early in the process, the team conducted a structured stakeholder analysis to identify additional participants with relevant expertise and decision-making roles. This approach evaluated stakeholder influence and interest to ensure inclusive and practical engagement (Brugha & Varvasovszky, 2000; Norström et al., 2020). Through this process, 10 collaborators were recruited to contribute to discussions, workshops, and review of outputs (Table 1). Hereafter, we refer collectively to the Co-PIs, TAG members, and collaborators as the *co-production team*.

Between October 2021 and March 2022, the lead PI and project coordinator conducted a targeted literature review summarizing primary threats to northern Gulf coast birds and evaluating stewardship and management actions used to address them. This review was shared with the co-production team prior to the first workshop to establish a shared understanding of threats, knowledge gaps, and uncertainties. Team members were invited to refine the review and contribute to its development as a manuscript (Fuller et al., 2025).

Using insights from the literature review and stakeholder input, the Co-PIs developed a problem statement and objectives to guide the SDM process, following the PrOACT framework (Gregory et al., 2012). The problem statement clarified decision context, management needs, constraints, and uncertainties (Smith, 2020), and was shared with the co-production team alongside the literature review and regional avian monitoring guidance (Wilson et al., 2019; Fournier et al., 2023) prior to the first workshop.

Co-Production Problem Framing and Identification of Uncertainties

In March 2022, two professionally facilitated workshops were held for the co-production team. The workshop, entirely virtual due to maintaining Covid-19 pandemic precautions, was facilitated by a consultant from the Consensus Building Institute (Cambridge, MA, USA), who assisted with agenda development, planning, and facilitation. During the first workshop, the full team assessed the literature review white paper, the Gulf Avian Monitoring Network guidelines, and draft problem statement, which they then refined through small and large group discussions. Participants also collectively agreed upon a fundamental objective which would address the decline of coastal bird population. While broad, this statement accounted for variability in goals of individual participants and was inclusive to a large number of alternatives to prioritize. Workshop participants continued working in small groups to brainstorm and refine a list of uncertainties that have the greatest impact on our ability to manage coastal bird populations. Participants drew from uncertainties identified in the Gulf Avian Monitoring Network guidelines (Wilson et al., 2019), as well as additional uncertainties identified based on their experience and expertise with researching and managing Gulf Coast birds. Over the course of the first workshop this list was refined to seven uncertainties influencing coastal bird stewardship and management:

1) The relationship between stewardship activities and demographic parameters of breeding and non-breeding birds.
2) The degree of impact human disturbance has on bird disturbance by proximity.
3) The degree of variance stewardship effectiveness has due to timing around the breeding season.
4) The degree of additive, synergistic, or competitive impacts from stewardship activities.
5) The magnitude of tradeoffs of predator exclosures between positive effects on chicks and negative effects on adult survival.
6) How and to what degree coastal habitat management impacts avian body condition, productivity, survival, and overall population size.
7) Impact of beach activities (e.g., beach driving) on avian habitat use and quality during the breeding and non-breeding seasons.

Influence Diagrams & Alternative Development

Two SDM-trained co-PIs then led two separate breakout groups that each developed a conceptual model to visualize relationships between management actions and outcomes while identifying underlying uncertainties. These models, referred to as influence diagrams (Howard & Matheson, 2005; Robinson & Fuller, 2017), identify management alternatives within the PrOACT process (Gregory et al., 2012). The influence diagrams focused specifically on coastal stewardship and management techniques used to address underlying threats and influence population size vis-à-vis coastal birds (the selected objective metric). Development of influence diagrams followed these steps: (1) brainstorming relevant management actions, (2) sorting management actions by level of uncertainty and impact, (3) connecting actions to population size through ecological response nodes, and

(4) refining the diagram from step 3 based on actions with high impact and uncertainty defined in step 2. The breakout groups presented their influence diagrams to all workshop participants for further discussion. Based on the group discussion, the SDM leads and lead PI then integrated the two influence diagrams and presented the resulting single model to workshop participants. Further large and small group discussion occurred during which the influence diagram was refined and ultimately accepted by all participants.

These influence diagrams were then used to develop a set of management alternatives to be considered for inclusion in the subsequent research proposal (NOAA RESTORE FO, NOAA-NOS-NCCOS-2022–2007377). Participants returned to their original breakout groups to generate a set of testable mechanisms from their refined influence diagrams. Alternatives (i.e., management decisions) were framed mechanistically to step beyond general management themes and actions and focus on prioritizing specific topics for further study. During the second workshop, participants reviewed recurring themes, a synthesized influence diagram, and research questions. The set of mechanistic alternatives were refined by cross-referencing with the literature to ensure accuracy and iterating between small and full group discussions.

CVoI Elicitation and Alternative Prioritization

A total of 22 workshop participants attended at least one of four alternative scoring sessions. The first was held in March after alternatives were refined at the end of the second workshop, and three more were held in July 2022 to accommodate participant availability and expand the range of perspectives and expertise included among the pool of scorers. These scoring sessions used the CVoI method, during which participants were trained and then independently and anonymously provided rankings (defined below) for each alternative using an online form. Each session began with a project refresher, a training on CVoI scoring, and a review of each alternative; each participant was provided with the list of 13 alternatives and the standardized scoring rubric (Table 2). This step contributed to the evaluation of consequences and tradeoffs within the PrOACT process (Gregory et al., 2012).

Scoring followed the same methods described in Stantial et al. (2023) (Table 2). Magnitude of uncertainty was scored from 0 to 4: 0 represented low uncertainty, in which there is clear evidence to support each alternative's mechanism of impact on the objective, whereas 4 represented high uncertainty where there is little to no evidence to support the alternative's mechanism. Relevance to decision making was scored from 0 to 3: 0 represented low relevance as the action does not change depending on the alternative's mechanism

Table 2 Full description of Constructed Value of Information (CVoI) scores 0–4 for Magnitude of Uncertainty (i.e., impact on the objective outcome), Relevance (i.e., likelihood of altering decision making), and Reducibility (i.e., feasibility to measure), adapted from Rushing et al. (2020)

Score	Magnitude of Uncertainty	Relevance	Reducibility
0	Firm theoretical foundation and a large number of empirical studies (>3) that support theoretical predictions	Preferred management action will be favored regardless of whether mechanism is true	Data necessary to reduce uncertainty do not currently exist and will be prohibitively difficult/expensive to collect given current technologies
1	Firm theoretical foundation with robust empirical support; OR Large number (>3) of consistent empirical studies	Reducing uncertainty is predicted to improve management outcomes but range of outcomes will be swamped by natural variability and other uncertainties	Data to reduce uncertainty exist but only for a limited taxonomic, geographic, or temporal scope but cannot resolve the specific mechanisms; collection of additional data needed to discriminate among alternative mechanisms will be difficult/expensive or cannot be collected in timeframe relevant to decision
2	Firm theoretical foundation with moderate empirical support; OR Moderate number (2–3) of consistent empirical studies	Reducing uncertainty is predicted to improve management outcomes; range of outcomes will be small to moderate compared to natural variability and other uncertainties	Data to reduce uncertainty exist but only for a limited taxonomic, geographic, or temporal scope OR data only allow weak inference about mechanisms; collection of additional data needed to discriminate among alternative mechanisms is feasible given current technologies
3	Firm theoretical foundation with no empirical support; OR Small number (<2) of consistent empirical studies	Reducing uncertainty is predicted to improve management outcomes and range of outcomes will be same order of magnitude as natural variability and other uncertainties	Data to reduce uncertainty exist across a large taxonomic, geographic, or temporal scope AND credible inference can be made from these data
4	Large disagreement between theory and empirical studies; OR No theoretical basis and inconsistent empirical studies		

of impact; 3 represented a high influence on decision making as the action changes significantly depending on the mechanism. Reducibility of each outcome was scored from 0 to 3: 0 represented low likelihood of reducing uncertainty for the alternative; 3 represented a high likelihood to reduce uncertainty (Rushing et al., 2020).

After the four scoring sessions, CVoI (Magnitude x Relevance) was calculated for each score entry, and the average CVoI and Reducibility scores for each alternative were plotted ($n=13$). The two-dimensional plot was divided into four quadrats of low, medium, high, and highest priority using the average Reducibility score to define the y-intercept and CVoI score to define the x-intercept (Lawson et al., 2022; Stantial et al., 2023). The full team reviewed the results, including Pareto optimality (e.g., maximization of CVoI and Reducibility) and used this information to guide a discussion and to refine the 13 alternatives into two final focal research questions.

Results

Problem Statement and Objective Metrics

After the first workshop discussion and refinement, participants agreed upon the following final problem statement:

> *Each year, natural resource managers decide which stewardship and management techniques to implement at coastal sites across the northern Gulf Coast to support bird populations. Beach habitats, in particular, are subject to a variety of stressors, many of which require specialized management actions to promote positive conservation outcomes for birds ranging from beach renourishment and revegetation to nesting habitat exclosures. Ideally, such resource management decisions would be targeted to reduce or mitigate the site- and species-specific threats and conditions for the community of affected birds, but it can be difficult to assess the trade-offs present in coastal habitat management and stewardship for the coastal bird community. Furthermore, many uncertainties remain about how to effectively implement stewardship and management to protect coastal birds and which techniques are most effective. In order to stabilize or increase coastal bird populations along the northern Gulf Coast, we need to design a research program that identifies key information gaps in the effects of stewardship and habitat management. This plan should help us optimally allocate resources to maximize the effectiveness of individual management actions at a given site, as well as efficiently prioritize actions across multiple sites and species.*

The co-production team's goals included identifying how to optimally allocate stewardship resources (e.g., stewardship staff, materials including signs and symbolic fencing), maximize management effectiveness, efficiently prioritize actions across sites and species, and identify the suite of actions with the greatest impact. The team collectively chose "support bird populations by reducing the current steady decline of coastal bird populations along the Gulf Coast and beyond" as their fundamental objective. This metric was also flexible in that it accommodated a variety of means objectives (e.g., increasing reproductive success, reducing sublethal impacts on adults and/or chicks, reducing lethal impacts on adults and/or chicks) that may or may not be met by specific management actions (e.g., predator management, human-related disturbance management, beach restoration). Due to the limited timeline in which the problem statement was developed, some, but not all components described in Smith (2020) were fully addressed. The co-production team determined that natural resource managers were the decision makers and that the overarching decision context involved stabilizing or increasing bird populations along the northern Gulf Coast. The focal decision was defined as determining which stewardship and management techniques should be implemented at coastal sites across the northern Gulf Coast on an annual basis, with the action to be taken as designing a research program to identify key information gaps and enable optimal allocation of resources and management. Objectives were framed primarily around a single overarching objective (support coastal bird populations) and a related means objective (stabilize or increase population growth rates).

However, the following components were lacking in the following areas and likely would have strengthened the statement if better articulated. First, legal, financial, and/or political constraints were not defined. The problem statement also mentioned that a variety of management actions may be used and the difficulty of assessing tradeoffs, but it did not explicitly address how broad or complicated the decision was. Uncertainty was also mentioned in general terms, but its magnitude on decision-making was not addressed.

Influence Diagrams and Alternative Selection

The two initial influence diagrams created by the breakout groups linked 26 total management actions (i.e., alternatives) relevant to coastal birds in the Gulf Coast region to the fundamental objective (i.e., reduce current decline of bird populations). Both diagrams were synthesized into a final influence diagram by removing redundancies and grouping into broader overarching themes (Fig. 1).

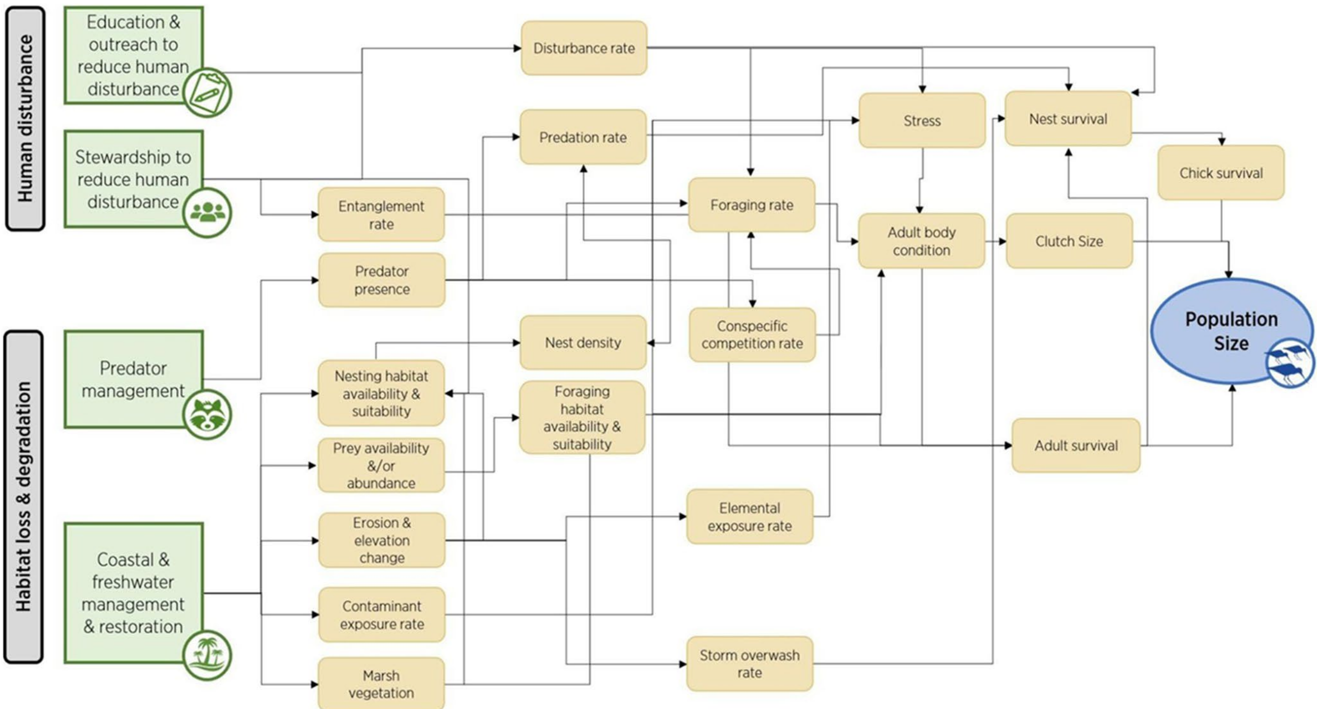

Fig. 1 Combined influence diagram linking the relationships between four total management actions (green) including human disturbance (top) and habitat loss/degradation (bottom), ecological processes (tan), and population size of beach nesting birds on the Gulf Coast (blue).(Reprinted from Fuller et al., 2025)

The co-production team identified three recurring management themes from the influence diagram created in the first workshop: (1) predator management, (2) habitat management, and (3) stewardship and outreach. The three themes also coincided with primary threats identified through the literature review (predators, habitat loss and/or degradation, human disturbance). Within each theme, the team identified three to six unique management actions to define alternatives (Table 3).

Table 3 Management themes and associated actions (or alternatives) derived from the influence diagram linking management actions to beach nesting bird population responses shown in Fig. 1

Management Theme	Management Action (Alternative)
Predator management	Nest exclosures
	Invasive and native predator removal and deterrence
	Other predator management techniques
Habitat management	Freshwater management
	Beach renourishment
	Restoration
	Back marsh creation
	Limiting beach grooming
	Other habitat management techniques
Stewardship and outreach	Leash laws
	Education and outreach
	Beach access policies
	Human activity management

Using the combined influence diagram (Fig. 1) and linked, recurring management actions (Table 3), 13 testable alternatives with proposed mechanisms of impacts were drafted, cross-checked with literature, and refined for scoring (Table 3). As previously mentioned, refining alternatives with the addition of mechanisms took a step further from broad management themes to create concrete research questions for testing in a subsequent study.

Alternative Scoring and CVoI Analysis

A total of 22 participants across four sessions scored each alternative for Magnitude, Relevance, and Reducibility according to the standardized rubric (Table 2). The average scores were then calculated to evaluate CVoI. The alternative with the highest Magnitude of uncertainty was 13) Stewardship sign design (2.87), and the lowest uncertainty was 10) Freshwater inputs (1.81). The alternative with the highest Relevance to decision making was 11) Allocating stewardship resources (1.96) and the lowest was 8) Nesting substrate (1.23). The highest Reducibility was 11) Allocating stewardship resources (2.13) and the lowest was 10) Freshwater inputs (1.33).

Alternatives were categorized under Low, Medium, High, or Highest priority based on their relative CVoI and Reducibility scores. Four alternatives were categorized as Highest Priority: 11) Allocating stewardship resources, 12)

Education and outreach, 6) Predators and vegetation, and 7) Beach renourishment (Fig. 2). Of these, 11) and 12) clearly stood out as the Highest Priority alternatives. A single alternative fell under High Priority: 4) Predator management duration. Medium priority alternatives in descending order were 13) Stewardship sign design, 5) Predator management efficacy, and (1) Nest exclosures. Low priority alternatives in descending order were 9) Beach height, (2) Lethal vs. non-lethal predator control, (3) Predator impacts by colony size, 8) Nesting substrate, and 10) Freshwater inputs.

Final Research Questions

The co-production team selected the following final research questions based on participant discussion and guided by CVoI analysis results (Fig. 2). Based on the highest priority alternative, 11) Allocating stewardship resources, the team refined the first research question as: "What is the relative efficacy of various stewardship techniques and intensities implemented during different phases of the breeding season and across a range of socio-ecological conditions?" From the second highest priority alternative, 12) Stewardship and outreach, the team refined the second research question as: "Are community-wide education and outreach campaigns effective supplements to on-the-ground stewardship to alter human behaviors?".

These research questions then served as the basis for a research proposal submitted to the subsequent NOAA RESTORE FO released in September 2022. All co-production team members were invited to join the research proposal as co-PIs or collaborators. Ultimately 12 co-PIs (including eight science producers and four science users) and 12 collaborators (including six science producers and six science users) worked together to prepare a proposal titled "Evaluating efficacy of stewardship actions for vulnerable Gulf Coast coastal birds through co-production between scientists and resource managers." This proposal was funded in 2023 (NA23NOS4510307) and co-produced research and dissemination of results to inform Gulf Coastal bird stewardship and management is ongoing through 2028.

Discussion

The efforts described in this paper demonstrated that co-produced research through SDM generated agreed-upon, high-priority research questions. Once answered, these research

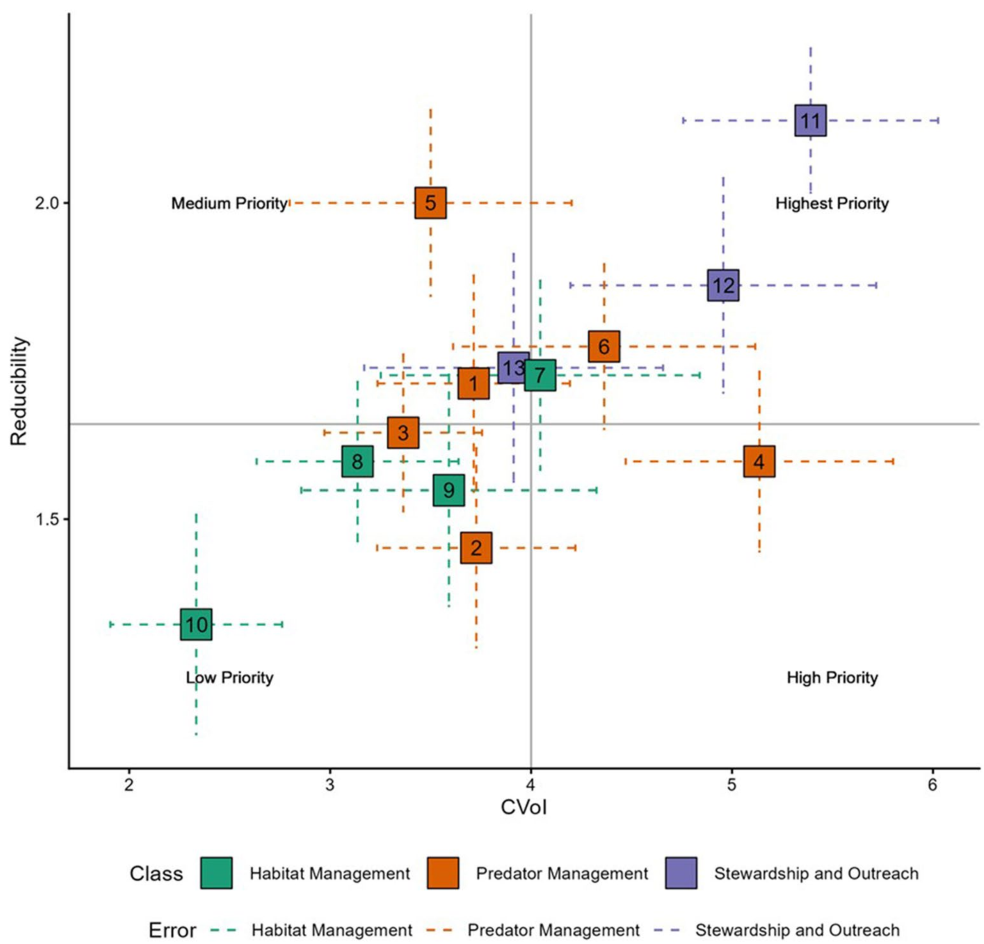

Fig. 2 Mean Reducibility (y axis) and Constructed Value of Information (CVoI) (i.e., Magnitude x Relevance) (x axis) scores plotted for each alternative from 22 scoring workshop participants. Colors represent recurring management themes: Habitat management (green), Predator management (orange) and Stewardship and outreach (purple). Dashed lines represent the uncertainty, or standard error (SE). The plot is divided into four quadrants, where the x-intercept represents the mean Reducibility score and the y-intercept represents the mean CVoI score across hypotheses. The highest priority category exhibits both high Reducibility and CVoI, high priority exhibits low Reducibility and high CVoI, medium priority exhibits high Reducibility and low CVoI, and low priority exhibits both low Reducibility and CVoI average scores

questions can guide future management efforts to conserve and stabilize bird populations along the Gulf Coast. While the inclusion of co-production concepts is not new to SDM (Runge et al., 2020; Nicholson et al. 2002; Starfield, 1997), social science literature has since expanded and formalized its definition and principles (Reed et al., 2014; Meadow et al., 2016; Beier et al., 2017; Djenontin & Meadow, 2018; Norström et al., 2020). Rather than asserting this work as a novel approach, we highlight the benefits of formally incorporating these tools and recognize their complementary qualities. Applying a co-produced SDM approach may be valuable in other settings where complex ecological and anthropogenic threats, numerous management alternatives, diverse interest parties and jurisdictions, as well as high uncertainty around management effectiveness exacerbate the decision-making process (e.g., Wilson et al., 2019; Fournier et al., 2020). Thus, such complex issues can benefit from broad expertise and perspectives to generate holistic approaches. Co-produced research through SDM enabled participants to understand, generate, and prioritize 26 possible management alternatives from diverse perspectives. The SDM PrOACT framework additionally provided the structure that enabled the full team to be involved throughout the process, summarizing and reducing uncertainty of the 26 management alternatives to produce 13 co-produced alternatives. As part of SDM, CVoI was key for selecting the two final research questions, as such a large number of hypotheses would have been otherwise too complex and unwieldy to prioritize.

Application of Co-Production Principles through SDM and PrOACT Frameworks

This case study demonstrates how SDM provided the structure for applying the co-production principles outlined in Norström et al. (2020) and translating them into actionable science, while the intentional application of formal co-production principles enhanced the effectiveness of the SDM process itself. The first principle of co-production (i.e., that it should be context-based; Norström et al., 2020) aims to establish the scope of the problem. This can include ecological, social, and economic factors, clarifying key concepts and terminology, and incorporating different needs and perspectives of diverse interest groups. The first step of SDM, problem framing, also includes this principle in that it encourages defining the context and developing a common understanding of the project scope. The co-production team achieved both a context-based approach and problem framing by establishing a shared knowledge base through the literature review (Fuller et al., 2025) and the Gulf Avian Monitoring Network guidelines (Wilson et al., 2019). They then drew upon both the literature and participants' expertise during group discussions to identify key uncertainties and develop a final problem statement. The diversity of participants' expertise and input also led to a more holistic and informed identification of key problems and objectives.

The second principle of co-production (i.e., that it should be pluralistic) highlights the need for incorporating a wide diversity of knowledge, such as disciplines, sectors, regions, and skillsets throughout the decision-making process. Similarly, SDM requires identifying who needs to be involved while defining the problem and decision context, ensuring that decision makers and those impacted by decisions are involved (Robinson & Fuller, 2017). Diverse participants and thus an enhanced knowledge base of an ecological problem (e.g., environmental, political, technical) and its regional variation improves SDM by ensuring the correct questions and ideas are brought to the table. For example, the inclusion of science users from across the Gulf Coast enabled the co-production team to understand the context-dependence of threats and the effectiveness of stewardship and management techniques to mitigate them, as both threats and strategies varied across the region. As a result, the team more efficiently generated applicable outcomes, delivering more unique contributions, and more holistically developing and refining each stage of PrOACT (Beier et al., 2017; Norström et al., 2020). We assembled a diverse representation of participants by employing a stakeholder analysis as recommended in Norström et al. (2020), which included 10 collaborators encompassing eight science users from four states and 11 different organizations. In addition, co-PIs and TAG members recruited three science producers from diverse disciplines (including migratory bird ecology, quantitative ecology, and avian behavior and field methods) in both academic and non-governmental organizations.

The third co-production principle (i.e., that it should be goal-oriented) emphasizes the importance of collectively identifying an overarching goal and recognizing multiple pathways to achieve it (Norström et al., 2020). The co-production team applied this principle by defining a shared, meaningful objective metric through SDM (i.e., supporting healthy coastal bird populations). They then identified a variety of ways to reach this objective by developing a set of alternatives through SDM and, through group discussions and influence diagrams, established how each alternative contributed to the common objective. Goal-setting was critical for establishing a common understanding of the problem, and participants' ability to determine final research questions that integrate the needs and values of impacted science users, producers, and other interest groups (Fournier et al., 2023). It furthermore enabled participants to focus their efforts effectively within a complex system (Runge, 2020) and provided a clear reference throughout the SDM process (Beier et al., 2017).

The fourth co-production principle (i.e., that it should be interactive) emphasizes active participation, sustained engagement, and learning through regular, facilitated meetings and workshops (Fournier et al., 2020; Norström et al., 2020; Wright et al., 2020; Saunders et al., 2021). Frequent communication is especially important for building trust, enhancing credibility, and increasing the likelihood that end-users will apply the knowledge generated (Norström et al., 2020). The co-production team was able to integrate this principle through SDM, which provided a guided structure that focused discussions and activities on achieving each PrOACT step in a collaborative setting. Regular meetings and third-party facilitation created multiple opportunities for engagement and feedback, including large-group and breakout discussions, virtual chat, email, and surveys. In addition, trained SDM coaches provided expertise and instruction that enabled participants to understand and fully participate in the process (Fournier et al., 2023). Regular meetings also likely helped maintain continuity and momentum, while the virtual format allowed more frequent scheduling and reduced costs, thus enabling participation despite team members' time constraints. Team members were invited to participate from the beginning of the SDM process through the final scoring sessions. All 36 members attended at least one workshop, 29 of whom were trained in SDM. This high level of engagement demonstrates how SDM can support co-production by incorporating diverse perspectives and facilitating the development of shared, high-priority research questions. Participation dropped during the CVoI phase, with 22 members (75%) contributing to the final scoring sessions. While the reasons for lower participation are uncertain, possible factors include scheduling conflicts, overlap with the start of a field season, reduced interest, or limited confidence in background knowledge. Future co-production processes may benefit from post-participation surveys to identify drivers of reduced engagement and provide recommendations for improving retention.

Application of CVoI in Structured Decision Making Frameworks

SDM and implementation through the PrOACT framework provided a transparent and equitable process for setting objectives and evaluating competing management alternatives, reframed further as research questions. By guiding participants through problem framing, objective setting, developing alternatives with refined mechanism, and evaluating consequences and tradeoffs, SDM enabled an efficient comparison of alternatives grounded in participant's needs, values, and constraints. CVoI gave us a framework for comparing different uncertainties around beach nesting birds based on end user and expert input. The results further highlighted the wide spectrum of priorities across management alternatives, which vary considerably depending on regional context, management uncertainties, and availability of resources (i.e., funding, personnel, and time). Additionally, including CVoI during the initial research proposal development phase as in this case study can identify projects that have a high likelihood to bridge the research-to-implementation gap. In this case study, stewardship and outreach actions consistently emerged with higher CVoI and Reducibility than predation and habitat management, suggesting this theme may provide greater returns under current uncertainty.

The literature review and participant consensus found that research measuring the impact of stewardship and outreach directly was not as extensive as research measuring predator or habitat management, particularly in the Gulf Coast region (Fuller et al., 2025). This is unsurprising because social science objectives are commonly underrepresented in conservation decision analyses in favor of ecological ones, due to limited expertise or comfort with social science methods (Niemiec et al., 2021; Robinson et al., 2019). Recognizing this oversight, we expanded beyond our ecological objective (e.g., population size) to also evaluate management alternatives (e.g., stewardship and outreach) and associated mechanisms that require implementation and assessment using social science-based methodologies. As a result, the management theme and its associated alternatives ranked considerably higher in terms of Magnitude, with alternative 11 exhibiting the highest Pareto optimality (Table 4; Fig. 2). This led to the selection of alternative 11) Allocating stewardship resources and 12) Education and outreach through the CVoI process (Table 5).

Both final research questions underscored the high value and uncertainty surrounding social science-based research questions. This reflects a broader consensus in the literature that significant barriers hinder integration with social science, despite its potential to substantially improve conservation management (Bennett et al., 2017; Dayer et al., 2020; Niemiec et al., 2021; Robinson et al., 2019). The inclusion of participatory modeling approaches, such as CVoI, improved the integration of social science in the decision making process (Robinson et al., 2019) and enabled science producers and users to better understand how pursuing different research hypotheses may impact their objectives (Gregory et al., 2012; Cleveland et al., 2024).

For example, when allocating stewardship resources (Hypothesis 11, Table 5), current management guidance generally recommends multi-species approaches to managing human disturbance and prioritizing high-use areas important to imperiled species (e.g., Florida Fish and Wildlife Conservation Commission, 2013; Comber, 2021). Yet, regional managers raised concerns about limited capacity to

Table 4 Reference number and name of 13 alternatives with their associated mean and standard error (SE) for Magnitude of Uncertainty (i.e., impact on the objective outcome), Relevance (i.e., likelihood of altering decision making), Reducibility (i.e., feasibility to measure), Constructed Value of Information (CVoI) (i.e., Magnitude x Relevance), and priority rank, calculated from 22 participant responses. Priority rank was assigned to each alternative based on where their plotted mean CVoI and Reducibility fell within four quadrants (Low, Medium, High, Highest) (Fig. 2)

Alternative no.	Alternative Name	Magnitude of Uncertainty (SE)	Relevance (SE)	Reducibility (SE)	CVoI (SE)	Priority Rank
1	Nest exclosures	1.86 (0.17)	1.95 (0.18)	1.71 (0.17)	3.71 (0.48)	Medium
2	Lethal vs. non-lethal predator control	2.32 (0.19)	1.55 (0.19)	1.45 (0.16)	3.73 (0.49)	Low
3	Predator impacts by colony size	2.23 (0.16)	1.64 (0.18)	1.64 (0.12)	3.36 (0.39)	Low
4	Predator management duration	2.55 (0.21)	1.91 (0.21)	1.59 (0.14)	5.14 (0.67)	High
5	Predator management efficacy	1.82 (0.25)	1.91 (0.2)	2 (0.15)	3.5 (0.7)	Medium
6	Predators and vegetation	2.64 (0.28)	1.59 (0.22)	1.77 (0.13)	4.36 (0.75)	Highest
7	Beach renourishment	2.59 (0.22)	1.59 (0.26)	1.73 (0.15)	4.05 (0.79)	Highest
8	Nesting substrate	2.5 (0.23)	1.23 (0.17)	1.59 (0.13)	3.14 (0.5)	Low
9	Beach height	2.05 (0.28)	1.59 (0.21)	1.55 (0.18)	3.59 (0.73)	Low
10	Freshwater inputs	1.81 (0.19)	1.38 (0.21)	1.33 (0.17)	2.33 (0.43)	Low
11	Allocating stewardship resources	2.74 (0.18)	1.96 (0.19)	2.13 (0.11)	5.39 (0.63)	Highest
12	Education and outreach	2.7 (0.21)	1.74 (0.24)	1.87 (0.17)	4.96 (0.76)	Highest
13	Stewardship sign design	2.87 (0.23)	1.39 (0.22)	1.74 (0.18)	3.91 (0.74)	Medium

apply stewardship effectively, and there was limited empirical evidence in the region and beyond guiding stewardship optimization across temporal and spatial scales. Resource managers expressed uncertainty about whether resources should be spread out across multiple colonies of varying sizes or concentrated to protect a few large colonies. However, the inclusion of science users in the co-production process ensured that issues such as these, which were less known among science producers, were raised and prioritized.

Another challenge to education and outreach (Hypothesis 12, Table 5) approaches can be seen in traditional education programs that might struggle to produce long-term behavioral change (Adelman et al., 2000; Rickinson, 2001; Schillerstrom, 2021). Emerging social marketing campaigns have demonstrated promise in a variety of conservation contexts (Eagle et al., 2016; Green et al., 2019; Abrams, 2020. While there has been extensive research conducted on stewardship impacts on human behavior along the Atlantic coast (e.g., Comber, 2021; Comber & Dayer, 2022; Dayer et al., 2022; Green et al., 2022), comparatively little has occurred along the Gulf Coast (Fuller et al., 2025), suggesting high potential to better understand the role and ability of education and outreach to influence conservation impact in the region.

In many real-world problems, management alternatives and research questions are often highly complex, and without participatory filtering, the resulting list can be unwieldy and may fail to fully account for stakeholder values and key uncertainties (Robinson & Fuller, 2017; Bolam et al., 2019; Rieder et al., 2021; Liberati et al., 2022). In our case study, applying CVoI allowed participants to collectively prioritize two out of 13 alternatives that would otherwise have been too complex and cumbersome to refine, and to agree upon them as the foundation for coordinated, large-scale research.

Value of Co-production and SDM in Avian Conservation

The loss of 2.9 billion birds across North America since 1970 highlights the relevance of effective, coordinated, and large-scale conservation efforts (Rosenberg et al., 2019). The recovery and conservation of declining species is particularly affected by multiple stressors and competing land-use objectives (Hodgson et al., 2009; Redpath et al., 2013), which is amplified when species, such as migratory birds, traverse large distances and encounter spatially diverse stressors. Ranges of avian species frequently cover large geographic regions and transcend jurisdictional boundaries. This is especially true of migratory birds, which face disproportionate population decline relative to non-migratory species in part due to greater risk of habitat loss and/or degradation to their breeding, stopover, and/or non-breeding grounds (Hardesty-Moore et al., 2018; Gilroy et al., 2016). Therefore, conservation efforts for such species can also span large, ecologically diverse, and multi-jurisdictional regions (Beier et al., 2017; Norström et al., 2020; Stantial et al., 2023; Grimm et al., 2025), with multi-jurisdictional collaboration and coordination. Like other wildlife conservation efforts, avian conservation is supported by rigorous, science-based solutions and thus can benefit significantly from large-scale coordination across science producers and users achieved through co-production and related processes (Saunders et al., 2021). As illustrated in the process described in this paper, SDM can provide a formalized framework in which diverse science producers, users, and other interest groups can identify objectives, evaluate trade-offs, and collectively prioritize transparent, evidence-based research questions to work towards species conservation, including those that cross jurisdictions and geographies, like many birds prioritized in avian conservation.

Table 5 Reference table of 13 alternatives with associated mechanisms of impact refined and reframed from an original set of 26 management alternatives. Alternatives are organized by their associated management theme shown in Tables 3, and given an associated reference number, reference name, and full description

Management Theme	Alternative no.	Alternative Name	Alternative Mechanism of Impact
Predator management	1	Nest exclosures	Nest exclosures have a positive impact on reproductive success by preventing predation of eggs. However, exclosures can cause nest abandonment by adults. We have evidence from a single species (Piping Plover, *Charadrius melodus*) in some regions that higher reproductive success with nest exclosures outweighs any loss from nest abandonment, but this needs to be investigated for specific species.
	2	Lethal vs. non-lethal predation control	Lethal mammalian (and possibly gull) predator control measures are more beneficial to reproductive success/survival than non-lethal mammalian predator control measures such as deterrents or exclosures, which reduce but do not eliminate predation pressure.
	3	Predation impacts by colony size	Removal of an individual predator would have a positive impact on reproductive success of colonial nesting birds more than a solitary nesting bird.
	4	Predation management duration	Birds exhibit site fidelity, returning to breed at sites where they were successful previously, while predators will return to sites with available prey indefinitely, necessitating ongoing predator management. If lethal predator removal increases reproductive success at a site, birds may return to that site the following year, and if predator removal is not continued, this may create an ecological trap if predators recolonize the site.
	5	Predation management efficacy	Predation management will be more effective in reducing predator densities and increasing bird reproductive success and survival at sites that are less accessible to predators (e.g., islands).
	6	Predation and vegetation	At the population level, increased nest predation due to greater vegetation density will exceed the benefits in juvenile survival fledglings receive from the additional vegetative cover.
Habitat management	7	Beach renourishment	Arthropod prey availability recovers in 1–3 months following beach renourishment, with an optimal grain size of 200–300 μm. Using the optimal grain size and timing renourishment projects in late fall - early winter (between fall and spring migration) will increase bird foraging success, body condition, and adult survival during breeding and non-breeding seasons.
	8	Nesting substrate	Adding stones to the beach substrate will increase nesting habitat quality and reproductive success by increasing nest camouflage and reducing Atlantic ghost crab (*Ocypode quadrata*) predation, benefits that will outweigh the potential negative impacts of reduced prey availability.
	9	Beach height	The benefits of increased beach height for coastal bird reproductive success show a unimodal response, with the greatest benefits at an intermediate height where nests are protected from overwash but chances of abandonment due to predators or weathering are low.
	10	Freshwater inputs	Optimal freshwater input regimes influence menhaden (*Brevoortia patronus*) abundance and thus impact productivity of Brown Pelicans (*Pelecanus occidentalis*) and other piscivores via nest provisioning rates. Similarly, freshwater input regimes influence oyster reef restoration and the wading birds that feed on them.
Stewardship & Outreach	11	Allocating stewardship resources	Concentrating stewardship resources at beaches with the largest focal species colonies has a larger beneficial effect on nest success and chick survival (breeding) or adult survival and body condition (non-breeding season) or site fidelity (either season) than rotating stewards among beaches regardless of colony size.
	12	Education and outreach	Supplementing active stewardship with social media-based education and outreach campaigns will lead to additive benefits in increasing nest success, chick and adult survival, body condition, and site fidelity. Impacts will be greater at beaches used primarily by locals, as well as on offshore islands visited by locals and without active stewardship.
	13	Stewardship sign design	Signs made by children have a greater impact on (a) community uptake (i.e., people noticing them), (b) number of people within closed area, and (c) bird metrics (nest success, chick/adult survival, site fidelity, body condition) than other types of signs (e.g., regulatory vs. advisory signs).

Conclusions

The U.S. Gulf Coast region exemplifies the complexity of threats, variation in interest group priorities, and uncertainty regarding effective management strategies that are common hurdles to avian conservation. This case study provides an empirical example of how co-produced research through SDM can transparently synthesize diverse knowledge, strategically prioritize alternatives, and identify research questions with the greatest perceived relevance to coastal bird conservation and potential to improve decision making by addressing key information gaps. Grounded in the perspectives of science producers and users from diverse geographies, backgrounds, organizations, and areas of expertise, this case study demonstrates growing evidence that co-produced research through SDM is a broadly applicable and effective approach for generating actionable science while navigating complex ecological challenges and diverse contexts, including wildlife conservation and management (e.g., Rushing et al., 2020; Lawson et al., 2022; Byrd et al., 2023; Fournier et al., 2023; Grimm et al., 2025; Sepulveda et al., 2022; Irwin et al., 2023; Doll et al., 2022; Regan et al., 2023; McEachran et al., 2024; Boehne et al., 2024). While this process effectively facilitated the development of priority research questions, evaluation of the extent to which implementation of this research influences and/or benefits management practices and outcomes is an opportunity for future research. Measuring the success of implementation through interest group engagement, impact on management decisions, and benefits to avian populations can help to understand the efficacy of co-produced research through SDM in advancing avian conservation.

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Data Availability This manuscript contains entirely original data which has not been submitted elsewhere.

Declarations

Ethics Approval Not applicable.

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